Project Proposal: Domain-Tuned Draft Models for Efficient Speculative Decoding in Code LLMs

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1. Problem Statement

Problem: The effectiveness of speculative decoding critically depends on alignment between the draft and target model token distributions; however, current approaches rely on generic draft models that fail to capture domain-specific patterns in code generation, resulting in low acceptance rates and under-realized inference speedups.

Relevance: Code LLMs power widely-used developer tools where inference latency directly impacts user experience and serving cost. While speculative decoding promises significant speedups, its practical effectiveness is limited by draft–target misalignment. Addressing this gap can unlock consistent, real-world efficiency gains without requiring major architectural changes.

Domain:

- *Inference Optimization:* Efficient LLM decoding strategies, including speculative decoding, token acceptance rate optimization, latency reduction (tokens/sec, TTFT), and memory-efficient serving (KV-cache utilization, batching).
- *Parameter-Efficient Fine-Tuning:* Adapting lightweight draft models using QLoRA/LoRA to align token distributions with target models while minimizing compute and memory overhead.
- *Evaluation:* Systematic benchmarking using code generation tasks, including accuracy (Pass@k), efficiency (acceptance rate, throughput), and resource

usage (GPU memory, compute cost), along with ablation studies to isolate the impact of domain alignment.

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2. Proposed Approach

Core idea: We propose to improve speculative decoding efficiency by aligning the draft model with the target model's domain-specific token distribution through parameter-efficient fine-tuning (QLoRA) on code data. This alignment is expected to increase token acceptance rates, thereby translating theoretical speedups into practical latency gains. We further evaluate this training-based approach against architectural alternatives such as Medusa and EAGLE-2 to understand the trade-offs between model adaptation and decoding-time optimization.

Project type: Working demo + Benchmark study

Models:

- Draft model: TinyLlama-1.1B (base → QLoRA fine-tuned on code)
- Main model: CodeLlama-7B-hf (4-bit quantized via bitsandbytes)
- Architectural baselines: Medusa (CodeLlama-7B + Medusa heads), EAGLE-2 (CodeLlama-7B + EAGLE-2 framework)

Datasets / benchmarks:

- Fine-tuning: CodeSearchNet (Python subset, ~400K functions), The Stack (Python, filtered)
- Evaluation: HumanEval (164 problems, Chen et al., 2021), MBPP (374 problems, Austin et al., 2021)

What is novel: This work introduces domain-aligned draft models for speculative decoding, shifting the focus from generic approximation to distribution-level alignment for improved acceptance rates. It provides the first systematic comparison between training-based alignment (QLoRA-tuned draft models) and architectural acceleration methods (Medusa, EAGLE-2) on code generation tasks, offering new insights into when model adaptation is more effective than decoding-time optimization.

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3. Related Work

#	Reference (title, authors, year)	How your project differs
1	"Accelerating Large Language Model Decoding with Speculative Sampling," Chen et al., 2023	Uses generic draft models; we introduce domain-tuned draft alignment to improve acceptance rates for code tasks
2	"Medusa: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads," Cai et al., 2024	Focuses on architectural multi-head decoding; we directly benchmark it against training-based draft alignment
3	"EAGLE-2: Faster Inference of Language Models with Dynamic Draft Trees," Li et al., 2024	Uses hidden-state extrapolation; we compare this representation-level approach against our QLoRA alignment approach on code benchmarks
4	"LoRA: Low-Rank Adaptation of Large Language Models," Hu et al., 2022	Introduces the PEFT technique we apply; our work uses QLoRA to align draft model token distributions, not for downstream task accuracy

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4. Expected Deliverables

- **Code repository:** QLoRA fine-tuning scripts (HuggingFace PEFT + bitsandbytes); speculative decoding pipeline with acceptance rate logging (vLLM); Medusa and EAGLE-2 integrations; unified benchmarking scripts across all four conditions.
- **Report:** Acceptance rate, latency, and throughput analysis across all conditions; Pass@k results on HumanEval and MBPP; ablation studies and failure mode analysis; discussion of trade-offs between training-based and architectural approaches.

- **Demo:** Recorded video — inference speed comparison across all four approaches on real coding prompts.

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5. Evaluation Plan

Accuracy: Pass@1 and Pass@10 on HumanEval (164 problems) and MBPP (374 problems) across all conditions — ensures speedup does not degrade output correctness.

Latency: Tokens per second and Time to First Token (TTFT), measured per condition on identical hardware (single NVIDIA T4/A100 GPU).

Resource usage: Peak GPU memory (GB) and KV-cache efficiency per condition, to assess deployment feasibility.

Ablation studies:

- *Generic vs domain-tuned draft:* Baseline speculative decoding (TinyLlama-1.1B generic) vs domain-tuned (QLoRA fine-tuned) — isolates the contribution of domain alignment.
- *Dataset size effect:* Draft fine-tuned on 10%, 50%, and 100% of the code corpus — measures whether acceptance rate scales with domain exposure.
- *Domain generalization:* Domain-tuned draft applied to non-code tasks (math, general QA) — confirms the improvement is domain-specific.
- *Method comparison:* All four conditions on a unified benchmark using the same main model and hardware — controlled apples-to-apples comparison.

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6. Timeline

Week	Plan
1	Literature review, environment setup (vLLM, HuggingFace, bitsandbytes), baseline speculative decoding with acceptance rate logging

Week	Plan
2–3	QLoRA fine-tuning of draft model on CodeSearchNet; domain-tuned experiments; compare acceptance rate and throughput vs baseline
3–4	Medusa and EAGLE-2 integration; full HumanEval and MBPP benchmark across all four conditions
4–5	Ablation studies, statistical analysis, failure mode investigation, report writing and demo recording